PHYSICAL AND CHEMICAL SCIENCE

Materials and Their Shapes

Natural and made materials, and changing shape by squashing, bending, twisting and stretching



wood wool plastic glass stretch

In this pack

Learn it: Materials and their shapes ★☆☆
Fact check: Sort the materials ★☆☆
Think it through: Which material? ★★☆
Investigate: Which band stretches most? ★★☆
Real-world case study: The potter of Rajshahi ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · C1 · MATERIALS

Materials and their shapes

★☆☆
Grade 1

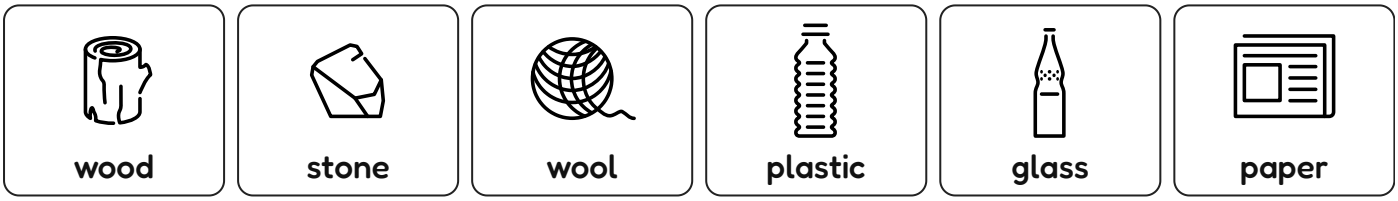
Name: Class: Date:

Learning goals

- Tell natural materials from made materials.
- Change the shape of materials.

Everything is made of **materials**. Some materials come from **nature**. Others are **made by people**.

Natural materials	Made by people
wood (from trees)	plastic
cotton and wool	glass
stone and clay	paper
jute	metal objects like spoons



Changing shape

We can change the shape of some materials by **squashing**, **bending**, **twisting** and **stretching**.

Action	Try it with
squash	a sponge, clay, play dough
bend	a pipe cleaner, a straw
twist	a towel, a rope
stretch	a rubber band, a balloon

Some materials go back to their shape (a sponge). Some stay in the new shape (clay).

Key words

natural	comes from nature
made by people	made in factories or by hand
squash	press hard
stretch	pull to make longer

FACT CHECK · C1 · MATERIALS

Sort the materials

★☆☆
Grade 1

Name: Class: Date:

Learning goals

- Sort natural and made materials.

1 Sort the materials.

SORT THESE

cotton wood plastic paper stone glass

Natural	Made by people

2 Match the action to the object.

Draw a line to match each pair, or write the letter.

squash ●	● **A** a sponge
stretch ●	● **B** a wet towel
twist ●	● **C** a rubber band
bend ●	● **D** a pipe cleaner

Which material?

★★☆
Grade 1

Name: Class: Date:

Learning goals

- Explain choices of materials.

1 Why is a ball made of rubber, not stone?

2 Which material goes back to its shape after squashing?

◯ clay ◯ a sponge

3 Draw something you can stretch.

INVESTIGATE · C1 · MATERIALS

Which band stretches most?

★★☆ Grade 1

Name: Class: Date:

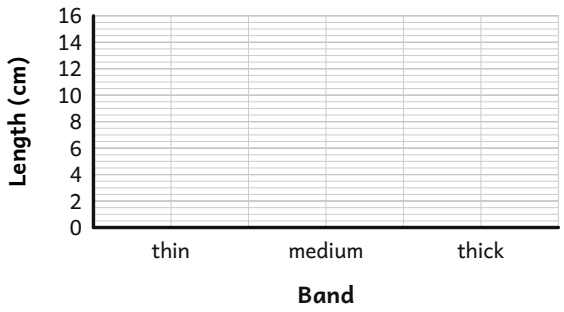
Learning goals

- Do a fair test.
- Read a bar chart.

Class 1 hung the same toy from three rubber bands and measured how long each band became.

1 Our results. Draw the bars.

Band	thin	medium	thick
Length (cm)	14	11	8



2 Which band stretched the most?

◯ thin ◯ medium

◯ thick

3 What did they keep the same?

REAL-WORLD CASE STUDY · C1 · MATERIALS

The potter of Rajshahi

★★★
Grade 1

Name: Class: Date:

Learning goals

- Read about a natural material.

Mr Paul is a potter. He digs **clay** from the river bank. He squashes and shapes the soft wet clay on his spinning wheel to make pots.

Then he dries the pots in the sun and heats them in a hot fire called a **kiln**. Now the clay is hard and will never go soft again.

1 **Is clay a natural material or made by people?**

◯ natural ◯ made by people

2 **How does Mr Paul change the shape of the clay?**

REVIEW AND CHALLENGE · C1 · MATERIALS

Show what you know

★★★
Grade 1

Name: Class: Date:

Learning goals

- Use material words.

1 Find the words.

S	P	C	J	R	M	B
M	T	W	I	S	T	O
G	L	A	S	S	F	C
F	F	G	S	V	T	L
E	W	O	O	D	M	A
H	A	K	C	J	M	Y
D	O	C	B	E	N	D

bend clay glass twist wood

2 Which is made by people?

◯ stone ◯ plastic

◯ wool

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
tell natural from made materials	◯	◯	◯
change the shape of materials	◯	◯	◯
test which band stretches most	◯	◯	◯